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Sushi Preference in Japan: A Discrete Choice Analysis

Understanding Consumer Behavior Through DCM — HW03

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What Drives Sushi Choice?

Research Question

- How do price, oiliness, and availability explain sushi preference?
- Do age and East/West region moderate taste for oiliness?
- Can we predict substitution & market scenarios?

Data: SUSHI Dataset

- 5,000 Japanese consumers (Kamishima, 2003)
- **Set A**: 10 items, full ranking
- **Set B**: 100 items, sampled sets
- Attributes: price, oiliness, freq. sold
- Demographics: age, gender, 12 regions

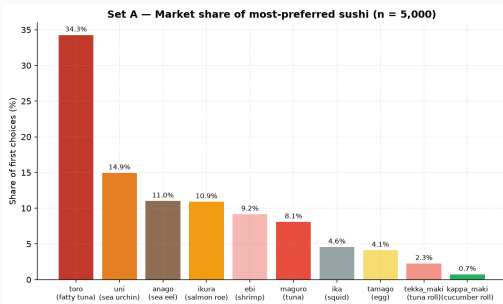


Figure 1: *

Toro dominates (34.3%). Uni, anago, ikura follow.

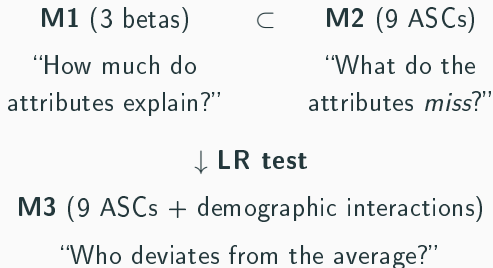
The Identification Challenge

Critical Design Property

Price, oiliness, and freq_sold are **constant per item** — all 5,000 users see the same values.

⇒ **ASCs and attribute betas cannot coexist** in one specification (perfect collinearity).

Solution: nested chain of models with LR tests



M3 — Preferred Model: Age Loves Oil

Specification

$$U_{nj} = ASC_j + \beta_{p \times w} \cdot price_j \cdot west_n + \beta_{o \times a} \cdot oil_j \cdot age_n + \beta_{o \times w} \cdot c$$

Interaction	Coef.	z
$\beta_{price \times west}$	-0.066	-1.46
$\beta_{oil \times age}$	+0.089	5.20
$\beta_{oil \times west}$	-0.058	-0.86

LL = -9,733 $\rho^2 = 0.155$ LR vs M2: $p = 10^{-9}$

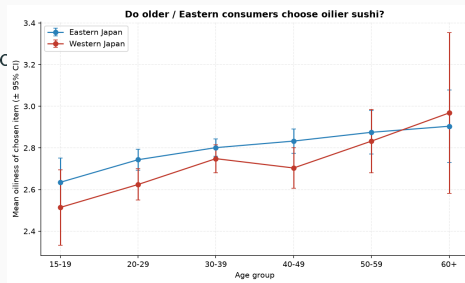


Figure 2: *

Older Eastern consumers choose the oiliest sushi.

Bottom line

Each step up in age group adds +0.089 utils per unit of oiliness — **older consumers consistently prefer richer, fattier sushi.**

East vs. West Japan: A Measurable Divide

Individual interactions: not significant

Both $\beta_{\text{price} \times \text{west}}$ and $\beta_{\text{oil} \times \text{west}}$ have $|z| < 1.96$.

But joint LR test: $p = 0.0004$

The East–West gap is *real* but price and oiliness are too collinear ($\rho = 0.82$) to separate cleanly.

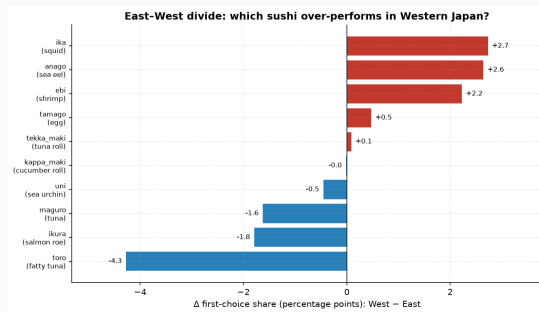


Figure 3: *

Systematic East–West differences in choice shares.

Why it matters

The README of the original dataset *predicts* this East–West divide. Our model *measures* it, but the attributes' collinearity prevents pinpointing the exact channel (price sensitivity vs. oiliness aversion).

Beyond MNL: Nested Logit Rejects IIA

Exploded ranking (ranks 1–3)

Each user contributes 3 pseudo-choices:

- Stage 1: all 10 items available
- Stage 2: 9 items (rank 1 removed)
- Stage 3: 8 items (ranks 1–2 removed)

⇒ 15,000 pseudo-observations

Three nests by ingredient

1. akami: maguro, toro, tekka_maki
2. seafood_other: ebi, anago, ika, uni, ikura
3. non_seafood: tamago, kappa_maki

λ parameter	Estimate	t vs. 1
λ_{akami}	0.325	−26.1
$\lambda_{\text{seafood_other}}$	0.336	−12.6
$\lambda_{\text{non_seafood}}$	0.002	−115.7

LR vs MNL = 668.7, $p \approx 0$

$\lambda_{\text{non_seafood}}$ hits the $\lambda \rightarrow 0$ boundary (degenerate 2-item nest); predictions use the M4b variant ($\lambda_{ns} = 1$).

$\lambda < 1$ in every nest

Substitution is much stronger within nests than MNL predicts.

Three Latent Consumer Segments (LC-MNL, Apollo)

Latent Class MNL on the exploded ranking; class membership driven by age, region, gender, migration. BIC selects **3 classes** (LL $-28,514$ vs. $-29,446$ with 1 class).

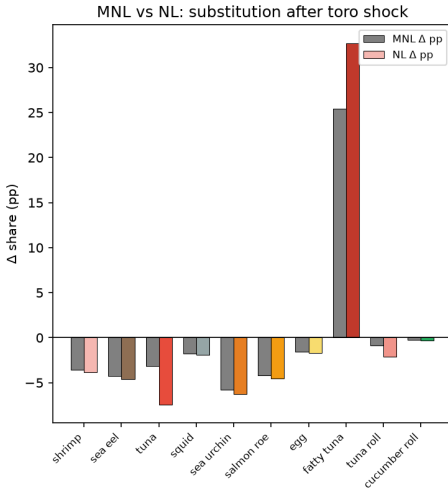
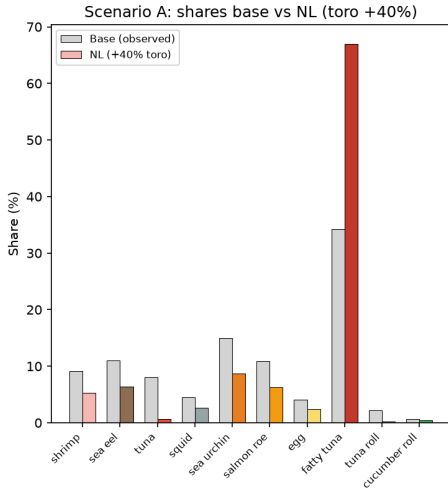
	Premium gourmet	Light taste	Tuna fanatic
Share	33.7%	31.3%	34.9%
Signature items	uni, ikura, toro	ebi (all else < 0)	toro, maguro, tekka
Rejects	egg, rolls	uni (-2.2)	cucumber roll
Profile	oldest, 51% F	youngest, West, 64% F	most male, East

Convergent evidence

The “tuna fanatic” class *is* the akami nest seen as a person type. And uni’s polarization (loved by gourmets at $+1.9$, rejected by the light class at -2.2) explains the rank-stability drift found in robustness checks.

Scenario A: What If Toro Becomes Scarce?

Intervention: Toro price +40% (bluefin tuna quota reduction)



Scenario C: Can an American Chain Enter Japan?

New item: California roll

- Price = 1.5, oiliness = 2.0
- Style = maki, type = seafood
- Model: M6 (100-item Set B)

Result	Value
Predicted share	1.50%
Sensitivity range	0.91% – 2.44%

Sensitivity: ± 0.5 on unobserved quality constant.

Top cannibalised items

Item	Loss (pp)
Toro	−0.14
Uni	−0.08
Ikura	−0.07
Anago	−0.07
Maguro	−0.07

Cannibalization proportional to shares (IIA).

Caveat: cannot assert “cannibalizes maki” without nest info for the hypothetical item.

Market insight

1.5% is non-trivial in a 100-item market. But the IIA assumption means we predict proportional draw from all items — a richer model is needed to test whether it would specifically cannibalize other maki rolls.

Strategic Recommendations

For Industry

1. **Menu design:** When toro sells out, stock up on maguro and tekka_maki — not proportional restocking.
2. **Regional targeting:** Premium fatty items → older Eastern consumers. Light items → younger Western consumers.
3. **Market entry:** A California roll would capture $\approx 1.5\%$ share. Non-trivial for one new SKU.

For Public Policy

1. **Don't subsidize healthy sushi:**
 $\beta_{\text{price}} > 0$ means cheaper = lower perceived quality.
2. **Boost availability instead:**
 $\beta_{\text{freq_sold}} = +2.17$ is the largest coefficient. Make healthier options visible.
3. **Reframe, don't discount:** Market light sushi as premium craftsmanship, not “cheap healthy alternative.”
4. **Cultural preservation:** Regional taste differences are real — leverage identity, not price.

Limitations & Next Steps

Key Caveats

- $\beta_{\text{price}} > 0$: price signals quality in this survey; WTP and price scenarios are **not monetary**.
- $\text{corr}(\text{price}, \text{oiliness}) = 0.82$ in Set A: multicollinearity inflates SEs for West interactions.
- Rank explosion assumes stable preferences across ranking stages.
- Set A is only 10 items; Set B extends to 100 but lacks ASC identification.

Validation & future work

- Apollo (R) cross-validation **completed**: log-likelihoods match the Python engine to ≤ 0.01 across all models.
- Latent Class (3 segments) estimated; a continuous Mixed Logit could capture residual within-class heterogeneity.
- Revealed-preference data for interpretable price effects.

Thank you